Keenan Heinz

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Education

University of Pennsylvania, School of Engineering and Applied Sciences Philadelphia, PA
Bachelor of Science in Engineering, Mechanical Engineering (Structures & Materials) *Aug. 2024 – May 2028*
Relevant Coursework: Intro to Mechatronics, Solids, Dynamics, Thermodynamics I, Analytical Engineering Methods, Intro to Scientific Computing, Electromagnetics

Engineering Experience

Mechanical Engineering Intern Menlo Park, CA
Stealth Startup *June 2026 – August 2026*

Automated Test Rig Fleet (Mini-Instron DAQ Systems)

- Multiple 500+ line design spreadsheets covering design criteria, BOMs, and hand calculations.
- Ran fastener and structural calculations to a 2.0 yield safety factor across all fixture designs, rated to a 50 kg maximum working load (based on design criteria).
- Designed and built wiring and control stacks: STM32 MCU, precision 24-bit ADC, and 3.0 mV/V load-cell signal chain.
- Wrote SPI-based DAQ software for synchronized force, displacement, and time data capture across the fleet.
- Self-taught DFM practices to prepare designs for outsourced fabrication.

Non-Contact Optical Extensometer

- Calculated and validated necessary focal length, position, and FPS limits to capture necessary deformation characteristics.
- Developed vision-processing software with gradient descent for sub-pixel changes and camera-controller synchronization for accurate post-processing.

Coil Winding System (ITC-PLC-70)

- Designed custom bobbins and tensioning system for consistent wire feed across varying coil geometries.
- Built HMI-integrated software to set and adjust coil requirement parameters on the fly.
- Ran test winds to validate machine scalability for future production volume.

Co-Lead Engineer Philadelphia, PA
Penn Hyperloop *Sep. 2025 – Present*

- Designed an A36 steel propulsion structure for an in-house Tunnel Boring Machine, in Fusion 360, rated for vertical loads >2500 N and horizontal loads >5000 N.
- Ran FEA and analytical hand calculations to validate structural integrity under worst-case loading.
- Evaluated 50+ suppliers for actuator sourcing as part of a full system redesign, achieving 150 kN thrust and 1+ meter stroke targets while minimizing procurement cost.
- Promoted from Propulsion Engineer to Co-Lead Engineer after two academic semesters; now directs mechanical and electrical integration across sub-teams for the 2027 competition machine.

Athletics

D1 and National Team Athlete, Rowing Philadelphia, PA
University of Pennsylvania / USRowing National Team *Aug. 2024 – Present*

- Two-time National Championship qualifier: 8th, 2026 IRA; 9th, 2025 IRA (team best points finish).
- 2nd, 2024 World Rowing Championships (St. Catharine's); 5th, 2023 World Rowing Championships (Paris).

Technical Skills

CAD & Simulation: SolidWorks, Fusion 360, Onshape, Finite Element Analysis, DraftSight, MATLAB
Electrical & Controls: wiring, signal chains, STM32, stepper drivers, ADCs, load cells (AI-assisted)
Manufacturing: DFM, production engineering (self-taught), MIG welding, ironworker, band saws, drill press
Programming: Python, Java, Excel, spreadsheet-based engineering calculations
Design: DaVinci Resolve, CapCut, Adobe Photoshop, Pixlr, PowerPoint, Canva
Interests: backpacking and camping, road biking, classic rally homologation cars, rock music, cats